Unsquared factors and linear selectors for Diophantine consistency

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Abstract

We give new explicit polynomial equations whose unsolvability in their stated domains is equivalent in ZFC to syntactic consistency of ZFC. The main natural-number construction evaluates an unsquared product of proof-checking factors instead of separately squaring every factor. A canonical CRT choice makes all newly exposed factors nonnegative, while preserving the semantic anchor and the exact source polynomial. The unrestricted-integer construction combines a signed radix with a linear binomial selector radius, removing redundant arithmetic witnesses. For ten positive variables, an inverse Lucas correction reuses existing registers but requires a new projection-first divisibility argument. A separate root-certificate circuit is shortened by exact identities and an integer coordinate shear. The release supplies literal equations, full fixed numerals, ordinary proofs, inverse maps, reproducible builders and independent checks. All statements retain their inherited calibration boundaries. No consistency decision, global minimum or new end-to-end kernel certificate is claimed.

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1 Objective and comparison scope

For each fixed polynomial $P \in \mathbb{Z}[x_1, \ldots, x_n]$ and specified domain D, the target is

$$\mathrm{ZFC} \vdash (\neg\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

We prove solvability equivalent to existence of a finite syntactic refutation. We do not prove either consistency or unsolvability separately. The domains $\mathbb{Z}$, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}_{>0}$ are part of the individual theorems and are not interchangeable complexity measures.

The comparison snapshot is `7a8da414376a80b90ecf799f37ceb26fb9ac674e` of ProveIt. It includes the new factorial-free, shared factorial-error, nearest-factorial, normalized-residue, digit-recovery and Lucas-correction incoming packages, as well as our preceding cofinal-witness report. We compare with their actual stronger equations, not only with our own older results. Incoming placement, repository adoption, ordinary proof review and fresh finite replay are recorded separately in the baseline audit.

The measurements are quantified coordinates, degree, quadratic residual count, fully collected support, maximum absolute coefficient and live binary evaluator operations. Addition, subtraction and multiplication cost one each, including squares and scalar products; register reuse is free. Fixed coefficient ports are not unknowns, but their entire coherent numerical tuples and finite definitions are part of every specification. Circuit degree and operation count do not measure expanded coefficient height or bit complexity.

1.1 Selected explicit equations

Domain and profile	Variables	Degree	Rows	Terms	Height
$\mathbb{N}$, dimension	206	4	158	8210	130686376
$\mathbb{N}$, balanced	207	4	159	5278	69238087
$\mathbb{N}$, sparse	208	4	160	5198	65866712
$\mathbb{Z}$, dimension	239	4	175	12658	6144
$\mathbb{Z}$, balanced	240	4	176	5232	4096
$\mathbb{Z}$, sparse	240	4	176	4927	7596

The natural 206-variable equation improves the matched preceding 219-variable/171-row/8267-term/height-130686428 equation in all four varying metrics. The sparse natural point similarly improves the preceding 221-variable/173-row/5253-term/height-65867916 equation. The middle point provides a separate dimension/support/height tradeoff.

The new 239-variable integer quartic improves the incoming shared-error 240-variable/177-row/21482-term/height-6144 point in dimension, rows and support at the same height. The sparse integer point improves the incoming factorial-free 242-variable/178-row/5018-term/height-7968 comparison in every varying metric. The balanced point trades support for a smaller coefficient height; these comparisons are not global variable-count records.

Domain and circuit	Variables	Exact degree	Binary operations
Strictly positive, inverse correction	10	304	85
Unrestricted, completed-square carrier	13	152	119

The positive circuit improves the incoming degree-304/87-operation Lucas-correction form at unchanged degree and domain. Its 84-operation/degree-410 square-weight companion is a dominated control. The 13-integer circuit improves the matched degree-152/126-operation incoming form. It remains a tradeoff against our earlier degree-156/ 117-operation point. Its coordinate shear is an exact integer bijection, not an identity in the original unshifted coordinate names.

Both numerical specializations of the positive circuit are supplied: the original radix67 tuple and the complete previously reviewed incoming complemented-label radix47 tuple. The latter has 207214 raw coefficient bytes instead of 295388; its encoding is credited prior work, not a new encoding theorem in this report. Counts and degrees are unchanged, while the specialized integer polynomials differ.

2 Canonical CRT signs and common-square extraction

Every coordinate in this section independently ranges over $\mathbb{N} = \{0, 1, 2, \ldots\}$. The selected ordinary quartics are $P(\mathbf{z}) = \sum_j R_j(\mathbf{z})^2$, with every quadratic residual and every expanded integer coefficient supplied literally in the packet.

Endpoint	Variables	Rows	Quartic terms	Height
Previous dimension control	219	171	8,267	130,686,428
Previous sparse control	221	173	5,253	65,867,916
New `natural206`	206	158	8,210	130,686,376
New `balanced207`	207	159	5,278	69,238,087
New `sparse208`	208	160	5,198	65,866,712

The new dimension and sparse equations improve their corresponding controls in all four displayed metrics; the 207-variable equation is a middle tradeoff. These are measured constructions, not claims of global optimality. Relative to the retained ordinary calibration from C222 through ZF to ZFC, each has a natural zero exactly when a finite syntactic ZFC refutation exists. Hence ZFC proves its no-natural-zero sentence equivalent to Con(ZFC). This is neither a consistency proof nor a newly kernel-checked end-to-end logical derivation. Factorials, Pell sequences and powers below belong to the ordinary proof, not to the syntax of the fixed released quartics.

2.1 An exact factorization of the unchanged local source

Use the old source's representative-minus-one coordinates a, b, c, d, S, S'. Write

$$\begin{aligned} \delta_0 &= S - S', & \delta_i &= S - S' - m_i \quad (1 \le i \le 8), \\ \delta_{11} &= (a-c)^2(b-c)^2 - (d+2)^2, & \delta_{12} &= S - S' - \texttt{mul.358}, \\ \delta_{13} &= \texttt{mul.363} - \texttt{mul.369}, & \delta_{14} &= \texttt{mul.372} - \texttt{mul.373}. \end{aligned}$$

Here $(m_1, \dots, m_8)$ are the already existing gates

$$(\texttt{mul.339}, \texttt{mul.341}, \texttt{mul.343}, \texttt{mul.345}, \texttt{mul.347}, \texttt{mul.349}, \texttt{mul.351}, \texttt{mul.353}).$$

The old axiom/rule disjunction is the literal nonnegative polynomial

$$R = \left(\prod_{i=0}^{8} \delta_i^2 \right) (\delta_{11}^2 + \delta_{12}^2) \delta_{13}^2 \delta_{14}^2. \tag{C1}$$

The complete local source adds the two unchanged beta-difference squares, $R + \beta_1^2 + \beta_2^2$. No change to the logical calculus is being made.

Define eleven oriented affine factors and two useful expressions by

$$g_i = -\delta_i \ (0 \le i \le 8), \qquad g_9 = -\delta_{13}, \qquad g_{10} = \delta_{14}, \qquad Q = \prod_{i=0}^{10} g_i, \quad E = (a-c)(b-c), \quad D = d+2.$$

Ten quadratic product rows construct Q successively, starting with $g_0 g_1$; affine factors require no new coordinate. One row constructs E. Three more quadratic rows construct

$$Z_0 = Q(E - D), \qquad Z = Z_0(E + D), \qquad T = -Q\delta_{12}. \tag{C2}$$

Replace the old evaluation input R by $Z^2 + T^2$, retaining the full source-evaluation row and its natural quotient. The integer identities

$$Q^2 = \left(\prod_{i=0}^{8} \delta_i^2 \right) \delta_{13}^2 \delta_{14}^2, \qquad (E - D)(E + D) = E^2 - D^2 = \delta_{11}$$

give $Z^2 + T^2 = R$, without any sign assumption. Thus *the entire local source polynomial is unchanged*, as are its majorant, prime-power decoding and logical interpretation.

The old fifteen square gates, two inequality-product gates and eleven disjunction-product gates become fourteen product gates: a saving of fourteen coordinates and fourteen rows before contraction. The new raw source has 310 natural coordinates and 262 quadratic rows. All thirteen nonanchor Pell seeds, eleven digit requests, six exact powers, source endpoints, semantic frame and derived label polynomial are retained.

2.2 Pointwise soundness and the genuine natural-domain issue

From any natural zero of the new source, restore each old square as its literal square and restore the old nonnegative product chain in order. These values are natural regardless of the signs of the affine factors. The identity above restores the old evaluation row; every other old row was retained verbatim. We have therefore obtained a natural zero of the complete old source, to which its established soundness proof applies.

Conversely, an arbitrary old tuple need not make the new partial products nonnegative. Consequently the transformation is **not claimed to be an orthant automorphism or a pointwise extension of all old witnesses**. The required completeness direction starts from a finite refutation and uses the freedom in choosing its canonical CRT representatives.

2.3 Cofinal sign choices preserve every CRT condition

First perform the parent's even descending construction, before choosing the encoded column representatives:

$$t = 2t_0 + 4, \quad F = t!, \quad n = F^2 - 1, \quad V = n^t, \quad y = V - 1.$$

Choose t larger than the finite proof-tree data and all its local labels. There are at most t pending nontruth vertices, each with weight at most $1+tF < n$, so every pending state is at most $(1+tF)^t < V$, hence at most y. Since t is even, $F \mid y$. The beta moduli $1 + (i+1)y$, $0 \le i \le t$, are pairwise coprime: a common prime would divide a positive index difference at most t, and thus F and y, contradicting its division of either modulus. Beta CRT gives x with the prescribed pending states and endpoint residues; enlarge x within its class as required. Define

$$B = \binom{n}{t}, \qquad N = B^x.$$

In particular, N has been fixed before any subsequent lift.

The inherited valuation construction uses positive representatives modulo N^2. Adding an arbitrary multiple of N^2 to any minus-one coordinate preserves its required residue, positivity and precision; there is no parity or square restriction on these natural coordinates. Choose the representatives for c, d, S, u, v and the encoded row index first. Next choose $b > c$ in its prescribed class and then $S' > S$ in its prescribed class. Finally choose a in its class so large that

$$\begin{aligned} a > c + d + 3, \qquad & a \ge \text{every previously fixed nonanchor index and coordinate,} \\ & (S+1)(1+F(a+1)) > (S'+1)(1+F(b+1)). \end{aligned} \tag{C3}$$

All these are lower bounds on a; a full tail of its arithmetic progression satisfies them. There is no dependence of N on these choices.

We now verify every new factor's sign. The factors $g_0 = S' - S$ and $g_i = S' - S + m_i$, $1 \le i \le 8$, are positive, as is $-\delta_{12} = S' - S + \texttt{mul.358}$. To check the two remaining factors, set $M = \texttt{mul.36}$. The literal inherited rows give

$$\begin{aligned} \texttt{mul.363} &= (S'+1)(1+F(b+1)), \\ \texttt{mul.368} &= (S+1)(1+F(a+1)), \\ \texttt{mul.369} &= (1+F(4+4b+2M))\texttt{mul.368}. \end{aligned}$$

Thus (C3) implies $g_9 = \texttt{mul.369} - \texttt{mul.363} > 0$. Also

$$\begin{aligned} \texttt{mul.372} &= (S'+1)(1+F(6+4b+2M)), \\ \texttt{mul.373} &= (S+1)(1+F(b+1)). \end{aligned}$$

Their inner weight difference is $F(5+3b+2M) > 0$; combined with $S' > S$, this gives $g_{10} > 0$. Finally $b - c \ge 1$ and $a - c > d + 3$ imply $E > D > 0$. Every partial product, Z_0, Z, T in (C2) is therefore natural.

The ordinary local residues are unchanged by these lifts, so the inherited precision argument still makes the nonnegative source a multiple of N. Its quotient is natural. This also holds for its exactly equal new expression.

2.4 The anchor is chosen after the signs, without a cycle

There is an important constraint beyond CRT residues: the semantic Pell anchor must dominate every requested seed index. It is not enough merely to enlarge the self-sized radix. Here the actual source anchor is unchanged:

$$J = 7 + 4b + 2M + \texttt{add.402}, \qquad M = (a+b)^2 + 5(a+b) + 6.$$

All remaining terms are nonnegative, so $J \geq 2(a+1)^2$. Condition (C3) makes every nonanchor seed index at most $a+1$, including the index $a+1$ itself. Hence

$$J \geq 2 \max_i k_i,$$

exactly the inherited sufficient condition for natural completeness of the one-row error-square cap. No lift of S or S' occurs afterward; neither is required to exceed J or a derived syntax weight.

Only now define the positive sum S_{base} of the independent representatives, with its fixed offset, and choose $q = (NS_{\mathrm{base}})^x$. Every representative is below S_{base}, whatever the size of its lift. The unchanged constructor ledger gives $J \leq S_{\mathrm{base}}^{\ell}$, $\ell = 2^{22}$. The inherited even-horizon bootstrap gives $y > 2^{36}$, $x > 2^{38}$, and $\ell(t+1) < x$. Thus all required nonfactorial low coefficients are at most S_{base}^x, and factorial low coefficients are at most $V < N$. The four pooled digit groups, of sizes $2, 5, 2, 2$, remain carry-free since

$$5S_{\mathrm{base}}^x + N < N^x S_{\mathrm{base}}^x = q.$$

Their low-remainder sums are at most one less than the respective denominator, so all pooled slacks are natural.

Compute the remaining denominator powers and frame W next; choose its lower Pell witness, a sufficiently large A in the common-root class, and then all principal, positive-center auxiliary, comparison and seed Pell witnesses. The anchor inequality supplies the natural merged-error quotient. None of these later choices changes the already fixed indices or CRT lifts. This completes the acyclic construction of a new natural zero.

2.5 Independent finite certification and artifacts

The source checker independently identifies all 28 removed gate definitions, all 14 new gate definitions, the evaluation-row replacement and all 247 otherwise unchanged rows. It verifies the factorization compositionally by exact integer identities, not by random evaluations. Corruption controls include deleting the source-evaluation row, changing a factor, widening the domain to integers and modifying an unrelated row.

The subsequent monic eliminations are separately and pointwise certified: every eliminated coordinate is an integer polynomial equal to a positive rational sum of affine squares plus nonnegative monomials. Independent replay proves their natural restorability and every substitution. They yield 206/158, 207/159 and 208/160 controls. Finally 25, 30 and 33 invertible integer *row* shears produce the selected endpoints in the table; coordinates and their natural domains do not change. Exact inverse operations, every expanded coefficient, refreshed metrics and printed residuals parsed back from TeX are verified both normally and with Python assertions disabled.

The full proof, complete literal quartics, pre-basis controls, certificates, offline rebuild and frozen ordinary arithmetic/logical closure are included under `natural/`. The new theorem uses canonical sign choices only for completeness; soundness and the later certified contractions remain pointwise. This distinction is essential to the reduction.

3 A linear binomial radius and a signed radix in one residue class

All equations in this section are ordinary quartics over independently unrestricted integers. They use the numerical C222 proof calculus directly, not a universal-polynomial machine-index substitution. Relative to the incoming factorial-free construction, two semantic edits eliminate two coordinates and two residuals:

$$R = F^2 \quad \longrightarrow \quad R = F, \qquad q = (2z+1)^2 \quad \longrightarrow \quad q = 4z+1, \qquad F = [2(t+1)]^t, \quad t = 16 + h^2.$$

Here the variable power describes the proved meaning of an unchanged quadratic Pell certificate; it is not an operation in the final quartic. The first edit uses a classical lcm estimate. The second allows an initially negative radix and then proves its positivity in the necessary order. Exact residual-basis changes subsequently improve the collected coefficients without changing the zero set.

Each final equation has the form

$$P(\boldsymbol{x}) = \sum_{i=0}^{m-1} R_i(\boldsymbol{x})^2 = 0, \qquad \boldsymbol{x} \in \mathbb{Z}^n,$$

with all R_i of degree at most two and P of exact degree four. Every residual and every collected coefficient is supplied, together with coordinate and inverse row certificates. The accompanying endpoint table records the measured tradeoffs. "Height" always means the largest absolute coefficient after complete collection of P.

Profile	Variables	Rows	Monomials	Height
Dimension	239	175	12,658	6,144
Balanced	240	176	5,232	4,096
Sparse	240	176	4,927	7,596

The 239-coordinate profile improves variables, residuals and collected support over the incoming Shared Factorial Error 240-coordinate endpoint, while retaining its height 6,144. All three new profiles have maximum residual coefficient 64. These are achieved tradeoffs, not global minima.

3.1 A classical lcm estimate makes the radius linear

Write $\Lambda_n = \operatorname{lcm}(1, \ldots, n)$. We recall the classical bound $\Lambda_n \le 4^n$, including a short proof so that the improvement does not depend on an unexplained asymptotic estimate. For $n \ge 2$, set $k = \lceil n/2 \rceil$. Then

$$\Lambda_n \mid \Lambda_k \binom{n}{k}. \tag{1}$$

Indeed, let p^a be the largest power of p not exceeding n. If $p^a \le k$, it already divides Λ_k. Otherwise $p^{a-1} \le k$, so only one further factor p is needed. The p^a term of Legendre's valuation formula for $\binom{n}{k}$ is one: n has one multiple of p^a, while k and $n-k$ have none. All other terms of that formula are nonnegative.

For $n = 2m$, the middle binomial coefficient is at most 4^m. For $n = 2m+1$, its two equal middle coefficients have sum at most 2^{2m+1}, so each is again at most 4^m. Complete induction in (1), starting with $\Lambda_1 = 1$, yields $\Lambda_n \le 4^n$ in both cases. This is a classical theorem, not a claimed new number-theoretic result; see Chan and Norrish, *Proof Pearl: Bounding Least Common Multiples*

with Triangles, J. Automated Reasoning 62 (2019), 171–192, Theorem 3, doi:10.1007/s10817-017-9438-0.

The unchanged horizon satisfies $t \geq 16$. Consequently

$$F = [2(t+1)]^t \geq 34^t > 16^t \geq \Lambda_t^2. \tag{2}$$

The authoritative factorial-free predecessor, its `report/main.tex`, proves the following two-sided binomial selector lemma under the hypothesis $R \geq \Lambda_t^2$: for $B = \binom{R+t}{t}$, every row $i \in \{1, \ldots, t\}$ has a prime power p^e exactly dividing B with $p^e > t$ and $p^e \mid R + i$; conversely, every prime power exactly dividing B divides $R + i$ for some row. The first assertion recovers every local proof row. The second handles *all* primes in the reverse CRT construction, including primes not selected to isolate a row. Equation (2) therefore permits $R = F$. An older predecessor review proves only a factorial-square threshold; that weaker review is not the theorem used here.

3.2 What is retained before signs are recovered

The complete signed squared frame, semantic anchor, pooled Pell orientation conditions, and all eighteen power slots are retained. Both coordinates of every principal Pell pair are oriented at their signed requested indices. No representative subset or frame atom is discarded. The inherited signed-power lemma, using the strict frame bound $|A| > W^W$, states that at a nonzero integer base b with $|b| \geq 2$, a requested index k must be nonnegative and its output equals b^k. For $b = \pm 1$, the output is 1 or $(-1)^k$ even for signed k. For a negative index at a nonunit base, the same small-error congruence would force an integer identity $mb^{-k} = 1$, which is impossible. We never apply this lemma to the base zero.

The known positive index t and base $2(t+1)$ first give $F = [2(t+1)]^t > 2^{32}$, independently of any radix sign. The retained beta certificate

$$v_\beta = (2w+1)^2, \qquad y = Fv_\beta$$

then yields $y \geq F > 0$. The possible boundary $y = F$ is not silently excluded; it is addressed after genuine local-row recovery below.

3.3 The mod-four radix and the acyclic bootstrap

The replacement $q = 4z + 1$ excludes both $q = 0$ and $q = -1$. It is stronger than oddness alone in precisely the needed respect: $q = -1$ would make the numerator base $q + 1$ zero. We now exclude the only remaining unit, $q = 1$. In that case the base $q + 1 = 2$ makes all nine actual CRT representative exponents nonnegative. Hence

$$H = 2 + \sum_\nu a_\nu \geq 2, \qquad C_0 \geq 0.$$

The retained raw index equation $B\mu = C_0 + R + 1$ has positive right-hand side, so $B \neq 0$ and $|BH| \geq 2$. The self-radix request gives $x \geq 0$ and $1 = (BH)^x$, forcing $x = 0$. But the last beta endpoint is

$$x = 1 + [1 + (t+1)y]u_t.$$

At $x = 0$ it would make the integer $1 + (t+1)y > 1$ divide -1. This contradiction uses neither a binomial interpretation nor a remainder bound.

For every remaining radix, put $Q = |q|$. Then $Q \geq 3$ and $|q + 1| \geq 2$. The same argument recovers nonnegative representatives, $H \geq 2$, and $B \neq 0$. The derived label is another requested

exponent at $q+1$, so it is nonnegative; the unchanged anchor excludes zero. The genuine self-radix equality is $q = (BH)^x$ with $x \geq 0$. Since $x = 0$ would give $q = 1$, we have $x \geq 1$. If the first beta quotient were $u_0 \leq -1$, its endpoint

$$x = F + 1 + (y+1)u_0$$

would imply $x \leq F - y \leq 0$. Thus $u_0 \geq 0$, $x \geq F + 1$, and the last endpoint has $u_t \geq 1$. It follows that

$$x \geq 2 + (t+1)y \geq 2 + (t+1)F, \qquad Q = |BH|^x \geq 2^x, \quad Q \geq 4B^2, \quad |B| \leq Q/4. \tag{3}$$

No sign of q or B has yet been assumed.

All powers at q and $q+1$ now have their genuine meanings at their nonnegative indices. In particular $M = q^x \neq 0$ and every digit denominator L_i is a nonzero power of q; its exponent is one of $t, t+1, y+1, x$, all at most x. For $L_i v_i = M$ and $v_i U_i = MQ_i + R_i$, the nonzero integer v_i divides R_i. Hence $r_i = R_i/v_i$ is an integer and $U_i = L_i Q_i + r_i$. The unchanged common norm

$$M^2 - 1 - 4\sum_i R_i^2 = a^2 + b^2 + c(c+1)$$

gives $|r_i| < |L_i|/2$ without assuming signs for L_i, v_i, M. Any congruent canonical tail of magnitude $< |L_i|/32$ must therefore coincide with r_i: their difference is a multiple of L_i of magnitude less than $(1/2 + 1/32)|L_i| < |L_i|$.

3.4 Authenticating the binomial at a signed radix

Put $n = R + t$ and $C_j = \binom{n}{j}$. The numerator request is exactly $U = (q+1)^n$. For $0 \leq j \leq t$,

$$C_j \leq (R+1)^t \leq (F^2+1)^t =: V.$$

The standard uniform binomial bound follows by Pascal induction and the last-edge product bound. Here $R = F \geq 1$; the mod-four-only controls instead retain $R = F^2$ and satisfy the same upper bound. The symbol V is proof notation, not a new power slot. For $F \geq 7$, $F^2 + 1 \leq 2^{F-1}$ (the base case and induction are immediate). Equation (3) gives

$$V/Q \leq 2^{-F-t-2}, \qquad V/(Q-1) < 1/32.$$

For either sign of q, the signed low polynomial

$$r^0 = \sum_{j=0}^{t-1} C_j q^j$$

has $|r^0| < Q^t/32$ and is congruent to U modulo q^t. The short-lift argument identifies the digit remainder with r^0. Dividing the digit equation by q^t and reducing modulo q now gives $B \equiv C_t \pmod{q}$. But $0 < C_t < Q/32$ and $|B| \leq Q/4$, so $|B - C_t| < Q$:

$$B = \binom{R+t}{t} > 0.$$

Since $H > 0$, the self-radix equality then proves $q = (BH)^x > 0$. The initially negative branch has been excluded by arithmetic. The request $N = B^x$ now has its ordinary positive nonunit base meaning. Also $B \geq R + t > R + 1$, so the raw index equation excludes $C_0 = 0$.

The remaining range and local-congruence arguments are unchanged. Their canonical tails are $< L_i/32$; all fixed front indices and local majorants are no larger after replacing F^2 by F. The strict endpoint budget $x \geq 2 + (t+1)F$ is preserved, including the derived-label bound. Applying the two-sided binomial selector theorem using (2) recovers every local C222 row.

3.5 Recovering the proof and constructing all witnesses

The recovered beta rows give accumulators $S_0, \dots, S_t$ and updates

$$S_{i+1}(1 + c_i F) = S_i(1 + p_i F)(1 + q_i F),$$

whose weights are positive. If $y = F$, the first endpoint would give $S_0 = 0$, while the last gives $S_t = 1$; every update propagates zero, a contradiction. Thus $y > F$, the initial remainder is genuinely $F + 1$, the final one is 1, and all accumulators are positive. Multiplication of the updates yields the inherited label-product identity. The two sides have equal nonunit factor counts $d \le t$: if they differed, the side with more factors would be at least $(F + 1)^{d+1}$, while the other is at most $t^d(F + 1)^d$, contradicting $F + 1 > t^t$. Each side is consequently the value at F of a product of at most t factors $1 + jX$, $1 \le j \le t$. Its coefficient sum is at most $(t + 1)^t < F$. Reduction modulo F at the least nonzero degree shows that a nonzero difference cannot vanish at F. The products are identical polynomials; their distinct roots $-1/j$ recover every label multiplicity. The retained Horn-closure argument therefore gives a finite C222 refutation and hence a finite ZFC refutation. This proves soundness.

Conversely, begin with a finite C222 refutation and choose a sufficiently large padded horizon $t = 16 + h^2$. Set $F = [2(t + 1)]^t$, $R = F$, and $B = \binom{F+t}{t}$. The two-sided selector theorem applies by (2). Let O be the odd part of $t!$. Choose an arbitrarily large odd multiple a_β of O, and put $y = Fa_\beta^2$. Since $v_2(F) \ge t > v_2(t!)$, this is an even multiple of $t!$ and can exceed F and every accumulator. The beta CRT works as before; its moduli are odd, so choose a sufficiently large even beta code x.

The smaller-radius binomial is still odd: $v_2(F + i) = v_2(i)$ for $1 \le i \le t$, since $v_2(i) < t \le v_2(F)$. Thus $v_2(B) = 0$ and $N^2 = B^{2x}$ is odd. The all-primes CRT construction provides the actual representatives. Adding N^2 permits their parities to be chosen while keeping the old bounds. Choose the dominant a odd and every other representative even; then a, b have the required opposite parities, the nonzero semantic anchor is even, and $H = 2 + \sum a_\nu$ is odd. The changed front index satisfies $R + t \le B < B^2 < N^2$, so every retained principal-index and frame bound remains valid. Now $q = (BH)^x$ is an odd square, hence $q \equiv 1 \pmod 4$, and $z = (q - 1)/4$ supplies the new affine radix constraint. Canonical normalized tails are $R_i = (M/L_i)r_i^0$ with $0 \le R_i < M/32$; the strict common norm gap has the usual ternary representation. The unchanged Pell completion supplies all other witnesses. Factorials are used only to select a divisible beta parameter, not as an authenticated unknown. This proves completeness.

3.6 Literal source edits, exact inverse certificates, and limits

In both pinned source equations $F = x_3$, $R = x_4$, and row zero is $R - F^2$. The source-interface audit checks that all previous factoring, projection, translation, and row-basis steps retain this row and these coordinates without using the row as a premise in an elimination. Replacing it by $R - F$ therefore implements exactly the semantic radius change. The centered stored radix has actual value $q = \text{stored}_q - 1$; its old row is $\text{stored}_q - 2 - 4z - 4z^2$. The new row is $\text{stored}_q - 2 - 4z$, so its affine restoration is $\text{stored}_q = 4z + 2$, not $4z + 1$. The second restoration is $R = F$.

These semantic edits preserve *zero existence*, by the two ordinary arithmetic directions above. They are not claimed to be bijections between all predecessor witness tuples. Only after the source edits do the two monic affine eliminations give exact restriction/restoration bijections. Every later row operation $R_i \leftarrow R_i + kR_j$ has the integer inverse with $-k$ and preserves the common zero set, though it generally changes the sum-of-squares polynomial.

The independent checkers import no new producer, verify all restored coordinates and pulled-back residuals, reverse every frozen row operation, collect every quartic coefficient, and check

degree, coordinate liveness, and all reported metrics. Deliberate corruptions must fail, under both normal Python and Python `-O`. These are finite artifact checks, not replacements for the unbounded arithmetic proof. The complete immediate predecessor, its ordinary proofs and calibration interface sources, is retained under `inherited_factorial_free`. The named logical interfaces are

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

Their composition yields the stated equivalence with Con(ZFC). No global minimum, proof of consistency, new proof-assistant closure, or literature-wide priority is claimed.

4 Inverse-Lucas correction: 85 operations in ten positive variables

Result and scope. The literal polynomial `positive/equation_bV.json` has ten independently strictly positive integer unknowns, exact degree 304, and 85 live binary arithmetic operations. The newest inspected Lucas-correction parent has the same degree and 87 operations. The square-weight control has 84 operations and degree 410; it is dominated by known earlier points and is not promoted as a frontier improvement. The complete indexed BFS943/radix67 tuple is unchanged, with largest fixed port of 592,884 bits.

Composed with the explicitly retained ordinary native/refutation calibration, the proof below gives

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in \mathbb{Z}_{>0}^{10}\ P(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con(ZFC)}.$$

This neither proves consistency nor asserts a zero. It is an ordinary mathematical proof with exact finite artifact checks, not a new end-to-end proof-assistant object or a global-minimum claim.

The new mechanism replaces the forward rank correction by an inverse Lucas correction at the existing abscissa k. This saves two multiplications and removes the exceptional negative-rank transport cases. A genuine obstruction at the prime 3 means that the old denominator proof cannot simply be reused. Projection must precede full denominator recovery; a new absolute bound handles a possibly negative corrected quotient in the positive lift. The second-order expansion and the projection-first strategy themselves are explicitly inherited from the inspected Lucas Corrections and Cofinal Witnesses archives at commit `7a8da414376a80b90ecf799f37ceb26fb9ac674e`.

4.1 Complete polynomial

The only unknowns are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The fixed integers β, K_0, M_C, M_F are the whole supplied tuple, with $\beta \geq 31$ odd, $K_0 > 0$, and $0 < M_C < \beta < M_F < 2\beta$. Define polynomial abbreviations:

$$\begin{gathered}
h = \beta J, \quad q = h + 1, \quad C = q - F - \alpha, \\
X = qw, \quad Y = qs, \quad k = \eta + \zeta, \quad c = kY + \eta, \\
A = Y(X + 1) + 1, \quad \Delta = A^2 - 1, \quad H = 4A - 5, \\
v = XY^2, \quad \ell = vk = (XY)(kY), \quad E = 1 + \ell(\ell + k), \\
R = [h(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J, \\
B = (K_0 + w)C - F, \quad U = ic^2, \quad a = \Delta U^2 E, \\
F_0 = 1 + a, \quad G = EF_0, \quad F_s = 1 + 4aF_0, \quad Q = \Delta(F_s - 1), \\
b = cT - R, \quad V = F_s b - c, \quad K = X(3c - 2X) + 2, \\
e_6 = 6(k - R - 1) - \ell(k^2 - 4), \quad D_6 = 6q^6, \quad M = hD_6H.
\end{gathered}$$

Use $L = bV$ for the principal equation, or $L = V^2$ for the control, and put

$$Z = hD_6\big[H(n - LB) + K\big] - H\big[D_6B + h(e_6 - 6q^4 s)\big], \tag{4}$$

$$P = Q(Z^2 - M^2GV^2) + (M^2G - Z^2). \tag{5}$$

The single equation is $P = 0$. All displayed powers have fixed nonnegative integer exponents. The DAG does not compute v separately: it reuses $\ell = (XY)(kY)$. There are no division gates, variable exponents, sequence functions, or additional quantified variables.

For proof only define

$$\begin{gathered}
e_* = \frac{e_6}{6} = k - R - 1 - \frac{vk(k^2 - 4)}{6}, \quad \rho = \frac{e_*}{q^6}, \quad \sigma = \frac{s}{q^2}, \\
t = \frac{B}{h}, \quad W = 1 + hL, \quad \kappa = \frac{K}{H}, \quad z = \frac{Z}{M}.
\end{gathered}$$

Without assuming any integrality of these quotients,

$$z = n + \kappa + \sigma - Wt - \rho. \tag{6}$$

4.2 Root extraction and uniform signed transport

The unchanged root and folding arguments are retained in `inherited_lucas/PROOF.md`, Sections 2–3, with their ordinary Pell sources. We state their premises and conclusions to specify the interface precisely. Positive inputs give $q \geq 32$, $X, Y \geq q$, $k \geq 2$, $c > k$, $A > 4$, $U \geq c^2$, $F_s > 2c$, and $Q > c^2$. Moreover

$$F_s = (1 + 2a)^2, \qquad Q = 4\Delta^2U^2G, \qquad \gcd(E, 1 + a) = 1.$$

At a zero, prime valuations in $(Q - 1)z^2 = G(QV^2 - 1)$ give $G = g^2$, $E = \tau^2$, $F_0 = f_0^2$, with $g = \tau f_0 > 0$ integral. Rational adjacent-Pell descent makes $y = |z|/g$ integral, and

$$QV^2 - (Q - 1)y^2 = 1, \quad z = \varepsilon g y, \quad \varepsilon \in \{1, -1\}.$$

Writing $f = 1 + 2a$, $I = 2Ug$, we have $f^2 - \Delta I^2 = 1$, $Q = (\Delta I)^2$, and $c^2 \mid I$. Signed all-index folding gives

$$c = \psi_A(p) \quad (p > 0 \text{ odd}), \qquad k = 2\psi_{1+2v}(m), \quad m \geq 2,$$

where $\psi_a(0) = 0$, $\psi_a(1) = 1$, and $\psi_a(j+2) = 2a\psi_a(j+1) - \psi_a(j)$. In particular $k \geq 8v + 4$, $c > 8XY^3 \geq 8q^4$, $c > X$, and $K > 0$.

The inherited exclusions of main and auxiliary index three use only $q \mid X, Y$, not the rank correction. They give, for either weight,

$$b \neq 0, \quad \operatorname{sign} V = \operatorname{sign} b, \quad |V| > |b|, \quad L > 0, \quad gy < L/2, \quad \frac{|b|}{q^6} < L/2. \tag{7}$$

For $L = bV$, the main index-three remainder would force $2Y - X \in \{1, 2\}$, impossible modulo q. The two signed auxiliary index-three possibilities would give

$$F_s(b - 4\Delta) = c - (4\Delta + 3), \qquad F_s(b + 4\Delta) = c + (4\Delta + 3),$$

whose right sides lie strictly between 0 and F_s. Thus the auxiliary index is at least five, giving $|b| > 4g$ and $gy < 3L/8$. Neither sign branch is discarded.

The new correction has an advantage before any sign of R is known: $k \geq 2$ makes $vk(k^2-4)/6 \geq 0$. Therefore

$$e_* \leq k - R - 1 < b, \qquad \rho < |b|/q^6.$$

If $B \leq -1$, then $Wt < -L$, and (6)–(7) force

$$n = \varepsilon gy - \kappa - \sigma + Wt + \rho < L/2 - L + |b|/q^6 < 0,$$

a contradiction. This handles all negative R and $R = 0$ uniformly. Hence $B \geq 0$, so $C > 0$. Only now is the inherited mask bound $3q < R < q^4 - 2$ invoked. Then $c > 2R$ and $b, V > 0$; the normalized-rank and signed auxiliary-index theorems give $p = R$. Thus R is positive odd; mask parity gives even q and odd J.

4.3 Inverse-Lucas authentication at all primes

The inherited, universally valid finite Lucas expansion gives

$$k = 2m + 8vC_m + 32v^2 Z_m, \quad C_m = \binom{m+1}{3}, \quad Z_m \in \mathbb{Z}_{\geq 0}. \tag{8}$$

Binomial notation occurs only in the proof. Since $k = 2u$ is even, $k(k^2-4)/6 = 4u(u-1)(u+1)/3$ is integral. Thus e_* is integral at a zero. Clearing the integral carrier gives $6q^6 \mid hH(e_6 - 6q^4 s)$; cancelling the ordinary factor six and using $h = q - 1$, $H \equiv -1 \pmod{q}$, gives

$$q^6 \mid e_* - q^4 s, \qquad q^4 \mid e_*. \tag{9}$$

No modular inverse of six is used.

Put $t_0 = 2m$, $d = 2m - R - 1$, $A_m = C_m + 4vZ_m$, and $S_m = k^2 + kt_0 + t_0^2 - 4$. The exact inverse identity is

$$3(e_* - d) = 4v^2(24Z_m - A_m S_m). \tag{10}$$

Indeed $t_0(t_0^2 - 4) = 48C_m$, $k - t_0 = 8vA_m$, and $k(k^2-4) - t_0(t_0^2-4) = (k - t_0)S_m$. Since $v = q^3 w s^2$, the right side is divisible by q^6. For each prime $\lambda \mid q$, with $a_\lambda \geq 1$ its valuation in q,

$$\operatorname{ord}_\lambda(e_* - d) \geq 6a_\lambda - \operatorname{ord}_\lambda(3) \geq 4a_\lambda.$$

Hence $q^4 \mid d$. The first base $1 + 2XY^2$ exceeds A. If $m \geq R$, monotonicity would give $k/2 \geq \psi_A(R) = c$, contrary to $c > k$. Thus $1 \leq m < R$, and $|d| < R < q^4$, proving

$$d = 0, \qquad m = (R+1)/2. \tag{11}$$

It would be false to infer full q^6-divisibility here. For $q = 3, v = 27, m = 2, R = 3$, the inverse error is $-47{,}911{,}824 \equiv 243 \pmod{729}$. This modular diagnostic is not a purported zero of the full polynomial. Evenness of q alone does not remove the obstruction at three.

4.4 Projection first, denominator second

Because $z = Z/M$ is integral, reducing the literal numerator (4) modulo H gives $H \mid 6hq^6K$. As $\gcd(q, H) = 1$,

$$H \mid 6hK. \tag{12}$$

This uses neither integral ρ nor integral σ, and does not cancel three. Write $R = 2r + 1$, so $r \geq 48$. The retained uniform ratio theorem, licensed by the authenticated indices, gives

$$Y > X^r/3, \qquad X^{r+1} < 3Y(X+1) < H.$$

Modulo odd H, the scaled recurrence gives $3 \cdot 2^{R-1}\psi_A(R) \equiv 4^R - 1$, because $4A \equiv 5$. Consequently $2^{R-1}K \equiv -(X - 2^R)(2^R X + 1) \pmod{H}$. Equation (12) therefore implies divisibility of this whole product by H, with multiplier $6h$.

For an explicit height check let $u_0 = 2^R = 2 \cdot 4^r$. Since $6 < q^3$ and $q^3 h < X^4$,

$$\begin{aligned} 6h\,|(X - u_0)(u_0X + 1)| &< X^4(X + u_0)(u_0X + 1) \\ &\leq 4 \cdot 4^r X^6 + 8 \cdot 16^r X^5 < X^{r+1} < H. \end{aligned}$$

After division by X^{r+1}, the two terms decrease with $X \geq 32, r \geq 48$; their sum at $(32, 48)$ is $2^{-117} + 2^{-25} < 1$. Hence the whole divisible product is zero, and its other factors are positive. Thus $X = 2^R$; the same universal projection then gives integral positive $\kappa = K/H$.

Now $q \mid X$ implies $q = 2^a$, $a \geq 5$. Equation (10), with $d = 0$, finally gives $q^6 \mid e_*$, because three is a unit modulo q. Thus ρ is integral. The carrier says $\sigma - WB/h$ is integral. Multiplying by q^2h and reducing modulo q^2 gives $q^2 \mid s$. Hence $q^3 \mid Y$, σ is integral, and $h \mid WB$. Since $W \equiv 1 \pmod{h}$, $h \mid B$, so t is a nonnegative integer.

The fixed-minus parity, exact half-binomial rounding, and fresh original native reconstruction now have their full retained premises. That source uses base $Y(X+1) + 2$, not the auxiliary base A. The unchanged whole-tuple calibration produces a finite ZFC refutation.

4.5 Constructive positive completeness

A finite refutation supplies the retained canonical outer data $q = 2^a$, $X = 2^R = qw$, $R > 3q$, $R \equiv 3 \pmod{4}$, $q^3 \mid Y$, $C > 0$, and $t = B/h \geq 0$ integral. Set $s = Y/q > 0$, and form

$$c = \psi_A(R), \quad k = 2\psi_{1+2XY^2}((R+1)/2), \quad \tau = \chi_{1+2XY^2}((R+1)/2), \quad \eta = c - kY, \quad \zeta = k - \eta.$$

The exact ratio theorem gives $0 < \eta < k$, hence both last inputs are positive. Equation (10) and dyadic q give integral ρ. Projection gives positive integral κ, and $\sigma = Y/q^3$ is integral.

A positive period of the Pell matrix modulo $c^2\tau$ gives a seed (f_0, I_0) with $c^2\tau \mid I_0$. Set $i = I_0/(c^2\tau) > 0$. Then $a = \Delta I_0^2 = f_0^2 - 1$, $F_0 = f_0^2$, and $G = (\tau f_0)^2$. The doubled pair $f = 2f_0^2 - 1$,

$I = 2f_0I_0$ has exactly the required definitions. The retained auxiliary-completeness theorem gives positive V, y and the positive integer $T = (V + c + Rf^2)/(cf^2)$, the actual unshifted input.

The new ρ may be negative; the old nonnegative-quotient argument is not reused. Since $k \geq R + 1$ and $vk^2 > 6$,

$$|\rho| \leq \frac{k - R - 1 + vk(k^2 - 4)/6}{q^6} < \frac{k + vk^3/6}{q^6} < vk^3/3.$$

For every permitted $i \geq 1$,

$$g > \sqrt{\Delta}\, c^2E > c^2v^2k^2 > vk^3,$$

so $|\rho| < g/3$. Also $\sqrt{\Delta} > A - 1 = Y(X + 1) > X$ and $cE > 4$, hence $g > 4Xc$. As $\kappa < Xc$ and $\sigma \leq Y < Xc$, we have $\kappa + \sigma < g/2$. For either positive weight put

$$n = gy + (1 + hL)t + \rho - \kappa - \sigma.$$

This is integral, and $y \geq 1$, $t \geq 0$ imply $n > g - g/3 - g/2 > 0$. Its carrier is $z = gy$, so $P = 0$. All ten independent coordinates have been constructed strictly positively; no inverse positive shear or unproved sign is used. Every permitted period seed works, without additional cofinal enlargement.

4.6 Exact costs and verification

Deleting the incoming gate $v = (XY)Y$, restoring the sharing $\ell = (XY)(kY)$, and replacing its four-gate rank cubic by k^2, $k^2 - 4$, $\ell(k^2 - 4)$ saves two multiplications. The principal has 51 multiplications, 18 additions and 16 subtractions. All inputs, ports and rows are live.

Put $q_0 = \beta J$, $k_0 = \eta + \zeta$, $C_0 = q_0 - F - \alpha$, and $B_0 = k_0sT - q_0^2(q_0 - F)$. The *complete* highest homogeneous forms are

$$[P_{bV}]_{304} = -9 \cdot 2^{12} q_0^{128} w^{34} s^{78} i^{14} k_0^{44} B_0^2,$$
$$[P_{V^2}]_{410} = 9 \cdot 2^{16} q_0^{166} w^{46} s^{104} i^{20} k_0^{60} B_0^4 C_0^2.$$

Their full expansions have respectively 274 and 3750 terms. Nonzero coefficients at maximal T-power establish the exact degrees for the actual $\beta > 0$, not merely a formal upper bound or a ray sample.

The producer-independent checker covers all actual rows with 37 exact free-polynomial cuts per graph, verifies full leading forms and every coefficient byte, reconstructs the four fixed numerals, and rejects both alternative operations at every gate plus a missing-correction mutation: 171 rejections for the principal and 169 for the control. Separate finite recurrence controls include the obstruction at three; they do not replace the unbounded arithmetic proof. Normal and optimized Python receipts agree byte for byte.

```
python3 -B positive/build.py --check
python3 -B positive/verify.py
python3 -O -B positive/verify.py
```

4.7 Whole complemented-label radix47 sidecar

The same two free-port row lists are separately specialized to the *entire* frozen complemented-label radix47 tuple. The principal remains 85/304, but its largest fixed port is 415,904 bits and the four raw numerals occupy 207,214 bytes rather than 295,388. This reuses the incoming encoding

and controller; it is not a new radix theorem and is not the repository's conditional three-anchor radix47 layout. The canonical radix67 equations remain separate.

The new arithmetic is uniform in the same scalar, source-shift mask and population interface already checked for the whole47 tuple. In soundness it recovers $C > 0$, $X = qw = 2^R$, $q^3 \mid Y$, the native half-binomial, and $(K_0 + w)C - F = t(q-1)$, $t \geq 0$. These are exactly the retained whole47 bridge's premises, with $W_{\text{native}} = w$ and source transport variable $t+1$. Its fresh Pell witnesses are constructed at $Y(X+1)+2$, before the actual complemented-label decoder is applied. Conversely, the actual whole47 controller supplies dyadic q and the same outer interface; choosing $w = W_{\text{native}}$, $s = Y/q$, and applying the explicit positive construction above gives a new zero. No old67 decoder or controller is silently substituted.

The complete ordinary native bridge, incoming P66 equation, all four payloads, 3297 coefficients, 51 native target identities, and its 37 pinned dependencies are retained under `positive/radix47/inherited/`. The sidecar checker requires identical free-port rows and an exact whole-tuple binding, reconstructs every numeral, and repeats the cuts, leading-form and mutation checks. Its normal and optimized receipts agree. The default main checker includes this sidecar; its literal binding can also be reproduced independently:

```
python3 -B positive/radix47/build.py --check
python3 -B positive/radix47/verify.py
```

5 Thirteen integers at degree 152 in 119 operations

The literal polynomial `integer13_degree152_ops119.json` has thirteen unrestricted integer inputs, exact degree 152, and 119 live binary operations: 71 multiplications, 24 additions and 24 subtractions. Squares and scalar multiplications are charged. This strictly improves the incoming 13/152/126 evaluator. Against the earlier 13/156/117, it trades two operations for four degrees. The new whole polynomial is exactly the incoming one under $n_{\text{old}} = n_{\text{new}} - k$; this is a polynomial automorphism, not a numerical or generic-parameter comparison. All six supplied coefficient integers are unchanged. Intermediate registers below are abbreviations, not unknowns.

The ordinary signed-index and native/compiler/refutation theorem is inherited from `roots/inherited/PROOF.md`, Sections 1–5, with its full retained source packet and finite numeral recipe. Thus ordinary ZFC proves that universal integer nonvanishing is equivalent to syntactic Con(ZFC). This does not assert consistency, global minimality, a new native encoding or a new end-to-end proof-assistant certificate. The incoming improvement from XY to XY^2 in the first-index modulus and from q^8 to q^6 in the budget is credited to that source, not to the present evaluator change.

5.1 The full division-free polynomial

The thirteen inputs, all in $\mathbb{Z}$, are

$$j, C, z, h, r, v, w, T, c, n, y, p_1, p_2.$$

The complete fixed ports are $\beta, K_0, M_C, M_F, A_c, B_c$, with M_F already shifted. In particular $\beta = 2^{1064921} - 1$, $A_c = 2^{1113}$ and $B_c = 2^{666104}$. Their full bytes are in the component's

`inherited/coefficients`, not unspecified parameters. Define

$$
\begin{gathered}
J = 6j + 1, \quad q_m = \beta J, \quad q = q_m + 1, \\
W = 12w + 4, \quad X = qW, \quad Y = v, \\
F = (K_0 + W)C - zq_m, R = [q(q - F) - C](q^2 - 1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
\ell = EY, k = R + 1 + h\ell, \eta = c - Yk, L = \ell k, \\
I = (2r + 1)c^2, D_s = \Delta I^2, F_s = 1 + D_s, Q = \Delta D_s, \\
V = F_s(cT - R) - c, \quad q_3 = q^3, M = 2q_3, U = MH, \\
K_{\text{lo}} = X(2X - 3c) - 2, D_{\text{err}} = (A_cF - q)^2 + (B_cC - q)^2, \\
B = (Yk)^2(q^2 - D_{\text{err}}) - q^2[4(Y\eta)^2 + (kq^2)^2], \\
G = 4(B - p_1^2 - p_2^2) - 3, D_t = 1 + L(L + k), \\
g_a = U^2G, d_b = 4U^2D_t, f_c = 16U^2F_s, \\
Z_c = U(n + 2L) + MK_{\text{lo}} + H(Y - q_3), \\
g_c = g_a + f_c, Z_2 = Z_c^2, g_{af} = g_af_c, \\
K = (Z_2 - g_c - d_b)^2 - 4(g_{af} + d_bg_c), \\
N = K^2 - 64g_{af}d_bZ_2, u_2 = (V + y)^2, \\
E_a = u_2 - Q(u_2 - V^2), P = (N + 1)E_a - 1.
\end{gathered}
$$

The auxiliary identity

$$
E_a = V^2 - (Q - 1)y(2V + y) \tag{13}
$$

is exact. Reusing u_2 saves one binary operation.

5.2 Whole-polynomial identity and inverse chart

Write, for proof only,

$$
Z = Un + MK_{\text{lo}} + H(Y - q_3), \quad b = 2L + k, \quad A(z) = (z + Uk)^2 + 4U(Lz - U).
$$

The incoming norm uses

$$
\begin{aligned}
K_{\text{old}} &= (A(Z_{\text{old}}) - g_a - f_c)^2 - 4[g_af_c + d_b(g_a + f_c)], \\
N_{\text{old}} &= K_{\text{old}}^2 - 64g_ad_bf_c[A(Z_{\text{old}}) + d_b].
\end{aligned}
$$

Under $n_{\text{old}} = n - k$ its numerator becomes $Z_{\text{old}} = Z - Uk$, and

$$
A(Z - Uk) = Z^2 + 4ULZ - 4U^2(1 + Lk), \tag{14}
$$

$$
A(Z - Uk) + d_b = (Z + 2UL)^2 = Z_c^2. \tag{15}
$$

Therefore $K = K_{\text{old}}$ and $N = N_{\text{old}}$ identically. Together with (13), this proves $P_{\text{new}}(n) = P_{\text{incoming}}(n - k)$ as whole polynomials. Since k is independent of n, the inverse is $n_{\text{new}} = n_{\text{old}} + k$ on every integer assignment. Only the degree-five shift k is removed; removing the degree-nine $2L$ as well would destroy the degree-152 cancellation.

5.3 All signed roots and all boundary cases

We give direct maps to the fifteen-input parent as an additional unbounded check. Unconditionally, q is even and nonzero, because $\beta > 1$ is odd and $\beta(6j+1) = -1$ is impossible. Thus $|q| \geq 2$, $|W| \geq 4$, $|X| \geq 8$, and X is even. Since $a = (X+1)Y$ is zero or has absolute value at least 7,

$$\Delta = (a+2)^2 - 1 \geq 3.$$

H is odd and nonzero, so $U \neq 0$. Also $\ell = XY^2$ is even, $\ell(\ell+1) \geq 0$, and

$$D_t = 1 + \ell(\ell+1)k^2 \geq 1 \text{ is odd}, \qquad F_s \geq 1.$$

$G \equiv 1 \pmod 4$ is odd and nonzero. These facts hold before any guard or zero assumption. Since N is a square modulo 4, $N \neq -2$. At $P = 0$ the integer factors $N+1, E_a$ are units; the first cannot be -1. Thus $N = 0, E_a = 1$, and the converse is immediate.

Put $\rho = Z_c/U \in \mathbb{Q}$. The universal complex identity is

$$\frac{N}{U^8} = \prod_{\varepsilon \in \{\pm 1\}^3} (\rho - \varepsilon_1\sqrt{G} - 2\varepsilon_2\sqrt{D_t} - 4\varepsilon_3\sqrt{F_s}). \tag{16}$$

It follows by multiplying all eight factors. If $G < 0$, its square root is the sole nonreal term of each factor, with nonzero imaginary part; therefore $N = 0$ is impossible. Hence $G > 0$. The case $G = 0$ is excluded by oddness; D_t, F_s are strictly positive, including their possible value 1.

Write the three radicands as $u_i^2 d_i$ with $u_i > 0$ integral and d_i squarefree. Distinct squarefree radicals, including $\sqrt{1}$, are independent over $\mathbb{Q}$: in the multiquadratic field generated by all their primes, character averaging of the sign-change automorphisms projects onto each radical. In any vanishing factor of (16), group terms by squarefree class. An irrational group containing G has an odd coefficient plus even coefficients. A group without G but containing D_t has a coefficient of exact 2-adic valuation one plus at most a multiple of four. A group containing only F_s is nonzero. None can cancel; u_i for G, D_t are odd. An even squarefree class can contain only F_s and is excluded likewise. This explicitly handles shared classes; no generic distinctness assumption is used.

Every class is therefore 1, and the signs of the zero factor give integers

$$g^2 = G, \quad \tau^2 = D_t, \quad f^2 = F_s, \qquad \rho = g + 2\tau + 4f. \tag{17}$$

In particular ρ is odd and nonzero. No division by ρ was used. All eight sign choices are allowed; f and τ were not assumed positive.

The fifteen-input kernel retains the common coordinates but has additional f, p_3, and residuals

$$A = (Z + Uk)^2 + 4U(LZ - U), \qquad B_s = f^2 - \Delta I^2 - 1,$$
$$C_a = V^2 - (Q-1)y(2V + y) - 1, \qquad G_b = B - 1 - p_1^2 - p_2^2 - p_3(p_3+1),$$

where its $Q = \Delta(f^2 - 1)$ and $V = f^2(cT - R) - c$. At a new zero restore

$$f_{\text{old}} = f, \quad p_{3,\text{old}} = (g-1)/2, \quad n_{\text{old}} = n - k - g - 4f.$$

All are integral. Its centered carrier satisfies

$$Z_{\text{old}}/U + (2L + k) = \rho - g - 4f = 2\tau.$$

The identity $A = (Z + U(2L+k))^2 - 4U^2 D_t$ gives $A = 0$. The equation $f^2 = F_s$ gives $B_s = 0$ and identifies both versions of Q, V; then $C_a = 0$. Finally $p_{3,\text{old}}(p_{3,\text{old}}+1) = (G-1)/4 = B - p_1^2 - p_2^2 - 1$, so $G_b = 0$. This is an actual parent zero.

Conversely, from any parent zero, $(Z_{\mathrm{old}}/U + 2L + k)^2 = 4D_t$. A rational square root of an integer is integral by reducing its fraction to lowest terms, and this root is even because its square is divisible by four. Write it 2τ. Set $g = 2p_{3,\mathrm{old}} + 1$ and

$$n_{\mathrm{new}} = n_{\mathrm{old}} + k + g + 4f_{\mathrm{old}}.$$

Delete $f_{\mathrm{old}}, p_{3,\mathrm{old}}$ and retain all other inputs. The old guard and strong row give $g^2 = G$ and $f_{\mathrm{old}}^2 = F_s$. The new carrier is $\rho = g + 2\tau + 4f_{\mathrm{old}}$, so a factor of (16) vanishes and $E_a = 1$. Hence $P = 0$. These constructions apply to every parent zero and add no new simultaneous Pell or canonical-witness existence claim.

5.4 Exact degree and complete leading cancellations

Naive max/sum degree propagation on the literal DAG gives only an upper bound 172. Both complete cancellation cuts below are essential. With all ports of degree zero, put

$$\begin{aligned}
q_0 &= 6\beta j, & X_0 &= 12q_0 w, & Y_0 &= v,\\
F_0 &= 12wC - q_0 z, & R_0 &= -q_0^3 F_0, & \ell_0 &= X_0 Y_0^2,\\
k_0 &= R_0 + h\ell_0, & a_0 &= X_0 Y_0, & U_0 &= 8q_0^3 a_0,\\
T_0 &= A_c^2 Y_0^2 F_0^2 + 4q_0^2 Y_0^4 + q_0^6.
\end{aligned}$$

The entire budget top is $[B]_{16} = -k_0^2 T_0$; in particular the $h\ell_0$ part of k_0 must not be dropped. The scaled radicands have degrees $28, 30, 24$. Although Z_c^2 has degree 30, the complete identity (14)–(15) shows that $Z_c^2 - d_b$ has degree 26, with top $-4U_0^2\ell_0 k_0^2$. Consequently the first square in K has degree 56, while $-4d_b g_a$ has degree 58, and

$$\begin{aligned}
[K]_{58} &= 64U_0^4\ell_0^2 k_0^4 T_0,\\
[N]_{116} &= 4096U_0^8\ell_0^4 k_0^8 T_0^2.
\end{aligned}$$

The outer subtracted product has degree 112 and cannot cancel this square. Independently, with formal weights $(Z, U, L, k, G, F_s) = (7, 6, 9, 5, 16, 12)$, the checker expands the entire norm and obtains the full top $256U^8L^4G^2$ of weight 116.

The second deliberate cancellation is $u_2 - V^2 = 2Vy + y^2$ of degree 18, although u_2 itself has degree 34. It gives

$$[E_a]_{36} = 32a_0^6 r^4 c^8 R_0 y.$$

The product of these displayed complete tops is the entire degree-152 output top. In particular, on the authentic coefficient slice,

$$[j^{90}w^{18}v^{22}z^9 r^4 c^8 y]P = 2^{167}3^{108}\beta^{90} > 0.$$

The exponents sum to 152. The symbolic extraction first sets $C = h = 0$, discarding only terms with variables absent from the target, and then selects the q_0^6 term in both copies of T_0. Thus the degree is attained, not merely a syntactic upper bound or a generic-port assertion.

5.5 Finite evidence and portable closure

The independent checker imports neither builder nor circuit implementation. It binds all cuts of the incoming 126-gate and new 119-gate DAGs, checks all topology, liveness, coverage, six fixed ports and costs, proves the whole shear and signed-factor identities, expands both leading cancellation

cuts, and extracts the stated actual coefficient. Each of the 119 live gates is separately corrupted and rejected. Normal and optimized Python runs both pass. All six raw coefficient payloads are reconstructed from the inspected retained finite recipe in both modes. The 188 retained files are SHA-256 pinned. These are checks of algebra and transcription, not substitutes for the unbounded proof above or for the inherited native/compiler theorem.

No twelve-variable breakthrough is claimed. Exploratory denominator-transfer, primitive-carrier and strong-scale variants did not produce a verified improvement and are not presented as theorems or new global obstructions.

6 Reproduction and mathematical boundaries

Every direct equation is the sum of the squares of a complete finite list of quadratic residuals. Both the residuals and every collected coefficient are supplied. The small-variable equations are complete straight-line DAGs over fixed integers and precisely the listed variables. No variable exponentiation, factorial, floor, binomial predicate or existentially unspecified constant remains in a final polynomial.

The portable top-level replay authenticates the complete release and runs normal and optimized Python checks in fresh disposable copies. It checks source edits, inverse/restoration maps, entire polynomial expansions, all circuit registers, exact leading forms, fixed numeral recipes and actual corruption controls. Finite testing is not used to replace any unbounded arithmetic theorem. The ordinary arguments establish soundness and completeness in the stated domains before the finite artifacts are used to bind those arguments to the literal equations.

The direct constructions retain the numerical C222 calculus and its ZF/ZFC refutation calibration, without K215 simulation or Jones universality. The compact and root branches have distinct inherited native calibrations and coefficient tuples. A whole alternate tuple changes the specialized polynomial; ports may never be mixed merely because they share names or satisfy some numerical inequalities.

There are different kinds of maps here. A polynomial triangular change of unrestricted integer coordinates is an actual lattice bijection. The natural source's new sign-oriented factors instead require a canonical CRT witness construction; arbitrary old natural witnesses need not lift pointwise. The factorial-free semantic changes likewise have separate unbounded soundness and completeness proofs. These statements are not conflated with the universal affine restoration certificates used later in compilation.

Reviews were performed separately within this AI-assisted workflow; they are not outside human endorsements. No upstream repository was modified. No global equation minimum, intrinsic arithmetic-complexity lower bound, literature-wide priority, new Lean/Rocq endpoint or encoded object-language ZFC derivation is asserted.

A Complete fixed-coefficient recipe

The following finite recipe is retained from the inspected ProveIt indexed constant specification, whose exact source and SHA-256 are recorded in `inherited_lucas/parent/proof_sources/indexed_constants.tex` and `SOURCE_PINS.json`. It defines the entire unchanged BFS943 tuple, not a new specialization. The symbol K in the inherited recipe is renamed K_0 to agree with Section 4; symbols in this section are local fixed data, not additional unknowns. The supplied standard-library reconstruction checks every resulting byte against both literal circuits. The source shift is already included in M_F. **The four fixed integers.** Every index, set, sum, product and

floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K_0 &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum_{i=0}^{942} v(t_i) x^i \\ &\quad + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} &= \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\ &\quad (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\ &\quad (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.\end{aligned}$$

The rational intermediate in K_0 cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$\begin{gathered}h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\}, \\ \mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\}, \\ r_q = 6 + |\mathcal{R} \cap \{0, \dots, q - 1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q - 1\}|.\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w, n, e, s)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j + 2\ (j = 2q + b < 430)$	$((1 - d_j) l_{u_j}, h(q, b), d_j r_{u_j}, v_j)$
$432 + b\ (b = 0, 1)$	$(0, h(215, b), 0, h(215, b))$
$434 + b\ (b = 0, 1)$	$(0, b, 0, b)$
$424 + 2r_q + b\ (q \in \mathcal{R}, b = 0, 1)$	$(r_q, b, 0, h(q, b))$
$424 + 2l_q + b\ (q \in \mathcal{L}, b = 0, 1)$	$(0, b, l_q, h(q, b))$
$924 + j\ (0 \le j < 4)$	$(0, j + 430\lfloor j/2 \rfloor, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$934+3a+b-6\lfloor a/4 \rfloor$ $a \in \{0,1,4\},\ b=0,1,2$	$(a, 437, b, 437)$

For $t = (w, n, e, s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

B Complete complemented-label radix47 constants

This is the entire incoming complemented-label tuple, not the repository's conditional three-anchor radix47 layout. All indices, labels and sums below are fixed data. The finite recipe and 436-character seed are reproduced from `radix47/inherited/provenance/radix47/radix47/equation.tex`; its K is renamed K_0. The independent reconstruction checks all 943 tiles, all 3297 coefficients, and every byte of all four actual fixed integers. No new encoding is claimed. **The four fixed integers.** Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{47}, \quad B = \rho^{8807}, \quad \beta = B - 1,$$
$$G_m(x) = \sum_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{4393} + \rho^{5495}.$$

$$\begin{aligned}
M_C &= \beta + \rho - (\rho+3)\rho^{1063} \\
&\quad - \sum_{e=0}^{3296} \lfloor (D_\rho(\Phi(\rho), e) + \rho - 1)/\rho \rfloor \, \rho^e, \\
M_F &= 2B + 3 + (2^{47} - 2)\Lambda \\
&\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\
&\quad - \rho^{8792} + (2^{28} - 3)\rho^{7694} - \rho^{7691}, \\
K_0 &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\
&\quad + \Lambda\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{aligned}
\Phi(x) = \sum_{i=0}^{942} &(1 + 2^{10} + 2^{10} w_i + 2^{18} n_i \\
&+ 2^{27} p_i + 2^{35} s_i) x^i \\
&+ 2^{10} x^{1003} (x^{20} + 2^{17}) G_8(2x) \\
&+ 2^{18} x^{983} (x^{30} + 2^{17}) G_9(2x) \\
&+ 2^{44} x^{992} (1 + x^{19} + x^{30} + x^{39}) \\
&+ 2^{44} x^{1095} (1 + x^{1100} + x^{2201}).
\end{aligned}$$

The rational intermediate in K_0 cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\},$$
$$r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \ldots, q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j+2\ (j = 2q+b < 430)$	$((1-d_j)r_{0,u_j}, h(q,b), d_j r_{1,u_j}, v_j)$
$434 - 2a + b\ (a, b = 0, 1)$	$(0, 432a+b, 0, 432a+b)$
$\substack{424+2r_{d,q}+b \\ q\in\mathcal{S}_d,\ d,b=0,1}$	$(dr_{d,q}, b, (1-d)r_{d,q}, h(q,b))$
$924 + 2a + b\ (a, b = 0, 1)$	$(0, 432a+b, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$934 + 3a + b\ (a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430+q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14 + m_q(D_{16}(E_0, \tau_q) - 14),$$
$$u_j = \nu_j(1+\sigma_j) + (1-\nu_j)D_{216}(T_0, j-\sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b+1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

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```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
OOe35uwDf15DpT6VJWZBjSbHPpPAINYTteOeUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```